Business Impact & ROI Model

AgriAdvisor Local Fallback Engine (Small-Scale Deployment Projection)

1. Baseline Metrics & Core Assumptions (Monthly Run-Rate)

- **System Drop Rate:** Approximately 250 queries drop per month due to rural network timeouts.
- **Manual Recovery Cost:** Support agents must call back farmers to resolve dropped queries.
- **Time & Labor:** Each manual callback takes an average of 12 minutes.
- **Labor Rate:** Fully loaded cost of a tier-1 support agent is ₹ 150 per hour.
- **Fallback Efficiency:** The local offline reasoning engine successfully answers 80% of dropped queries natively.
- **Customer Value:** Average Revenue Per User (ARPU) is ₹ 50/month. Unanswered drops have a 20% churn risk.

2. Operational Cost Savings (Time & Labor)

Metric	Calculation Logic	Impact
Total Manual Time Avoided	250 drops × 80% handled locally × 12 mins per resolution	**40 Hours / mo**
Labor Cost Reduced	40 hours saved × ₹ 150 per hour	**₹ 6,000 / mo**
API Costs Avoided	200 fallback queries rerouted from cloud LLM	**Negligible**

3. Revenue Retention & Risk Mitigation

Metric	Calculation Logic	Impact
Churn Events Prevented	200 local resolutions × 20% risk of churn if left unanswered	**40 Users Retained**
Monthly Revenue Protected	40 retained users × ₹ 50 ARPU	**₹ 2,000 / mo**

4. Total Quantified Business Impact

Total Monthly Impact	Projected Annual ROI	Support Capacity Freed
₹ 8,000	₹ 96,000	480 Hours/yr

5. Qualitative & Strategic Benefits

- **Farmer Trust & Reliability:** Providing instant answers even in zero-network zones builds high brand loyalty and reliance on the AgriAdvisor platform.
- **Support Team Mental Bandwidth:** Removes highly repetitive, low-tier follow-up calls from the support queue, allowing agents to focus on complex agronomy escalations.
- **Graceful Degradation:** System degrades from "Cloud Intelligence" to "Local Intelligence" rather than showing an error screen, vastly improving the UX.